Video-RAG: Visually-aligned Retrieval-Augmented Long Video Comprehension

Taro Yamada

Abstract

Understanding long videos remains a fundamental challenge for Large Video-Language Models (LVLMs) due to limited context windows and the visual redundancy introduced by dense frame sampling. Existing solutions—such as long-context fine-tuning or GPT-4-based agent frameworks—are expensive, data-hungry, and often rely on proprietary models. In this paper, we propose **Video-RAG**, a training-free, plug-and-play pipeline that augments any LVLM with visually-aligned retrieval-augmented generation. Instead of densely packing more visual frames, Video-RAG first decouples the user query into structured retrieval requests (ASR / OCR / object-type), builds three per-video text databases using open-source tools (EasyOCR [9], Whisper-large [25], and APE [29] paired with CLIP-ViT-L/14-336 [24] for keyframe selection), and retrieves query-relevant text snippets via Contriever [8] with FAISS [12] (threshold-based range search at t=0.3). The retrieved auxiliary texts are prepended to the query in DET→ASR→OCR order, boosting cross-modal alignment without any model retraining. Evaluated on Video-MME [6], MLVU [42], and LongVideoBench [35], Video-RAG delivers consistent gains across seven open-source LVLMs, achieving an average +2.8% on Video-MME with only ∼2.0K extra tokens per sample. With LLaVA-Video-72B [41], Video-RAG attains 77.4% on Video-MME (surpassing GPT-4o's 77.2%) and 65.4% on LongVideoBench (surpassing Gemini-1.5-Pro's 64.0%).

1 Introduction

Video understanding is a longstanding challenge at the intersection of computer vision and natural language processing. The emergence of Large Video-Language Models (LVLMs) [17, 40, 41, 33, 39] has dramatically advanced performance on short-video tasks, but long videos—those spanning minutes to hours—remain an open problem. Long videos pose two compounding difficulties: (1) the video content far exceeds the context window of current LVLMs, and (2) uniform frame sampling leads to severe redundancy, diluting the informative frames with uninformative ones.

Existing approaches to long video understanding fall into two broad categories, both with significant drawbacks (see Figure 1). *Long-context fine-tuning* methods [39, 37, 41] extend the model's context window at substantial training cost and are often tied to a specific architecture. *Agent-based methods* [4, 18, 19] decompose complex tasks into sub-queries handled by multiple specialized modules, but they typically rely on proprietary APIs (e.g., GPT-4V), incur high computational cost due to multi-turn inference, and lack interpretability.

A less-explored alternative is to enrich the *text* side of the multimodal input rather than the visual side. Videos are inherently rich in textual signals—spoken words (ASR), displayed text (OCR), and identifiable physical objects—all of which can be extracted using lightweight open-source tools. These modality-specific signals can serve as *auxiliary texts* that complement the visual frames, helping the LVLM reason across the video without requiring additional visual tokens. Crucially, this extraction is deterministic and model-independent, so the databases can be pre-computed once per video and cached for reuse across queries. The key challenge is then *selective* retrieval: identifying

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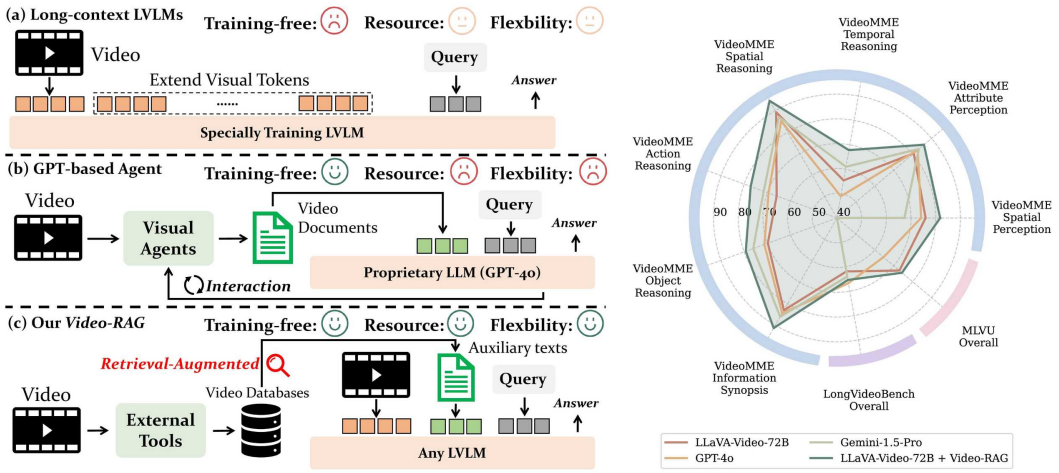

Figure 1: (Left) Two common approaches for understanding long videos: long-context fine-tuning and GPT-based agent pipelines. (Right) Our Video-RAG: a training-free, plug-and-play pipeline that augments any LVLM with visually-aligned auxiliary texts retrieved in a single forward pass.

which portions of these large text databases are most relevant to a given query—precisely the regime where RAG-based methods from the NLP community excel.

In this paper we introduce **Video-RAG**, a training-free, plug-and-play retrieval-augmented generation (RAG) pipeline for long video comprehension. Given a user query and a video, Video-RAG first prompts the LVLM itself (without any visual input) to produce a structured retrieval plan—specifying what ASR, OCR, and object detection information might help answer the question. It then retrieves the most relevant text snippets from pre-built per-video databases and prepends them to the final query. This single-turn, parallel retrieval process requires no model fine-tuning, introduces only $\sim$2.0K extra tokens on average, and is entirely built on open-source tools.

Our key contributions are:

- We introduce **Video-RAG**, a training-free, visually-aligned RAG pipeline for LVLMs that uses only open-source tools (EasyOCR, Whisper, APE, Contriever, FAISS).
- We design a plug-and-play, single-turn retrieval process that improves cross-modality alignment and reduces hallucinations, with minimal impact on inference efficiency.
- We demonstrate consistent gains across Video-MME, MLVU, and LongVideoBench on seven open-source LVLMs, with particularly strong improvements on long-duration videos.
- We achieve proprietary-level performance with open-source 72B LVLMs, surpassing Gemini-1.5-Pro and matching GPT-4o on multiple benchmarks.

2 Related Work

Large Video-Language Models. Early LVLMs for video understanding [22, 16, 17, 11] extended image-language models by aggregating frame features with temporal pooling. Subsequent work improved temporal reasoning through instruction tuning [40, 17], and recent models [41, 33, 3, 14] scale to billions of parameters, supporting richer video semantics. However, these models are fundamentally constrained by their context window: with fixed frame budgets (typically 32–128 frames), they either miss fine-grained temporal events or fail to retain sufficient global context.

Long Video Understanding. Several directions have been explored to handle long videos. Token compression methods [11, 31] reduce visual tokens via clustering or memory, at the cost of losing fine-grained information. Long-context models [39, 37, 41] extend context windows, but remain data and compute intensive. Agent-based approaches [4, 18, 19, 34] decompose complex queries into tool-calling pipelines; while powerful, they require multi-turn inference and often rely on closed-source

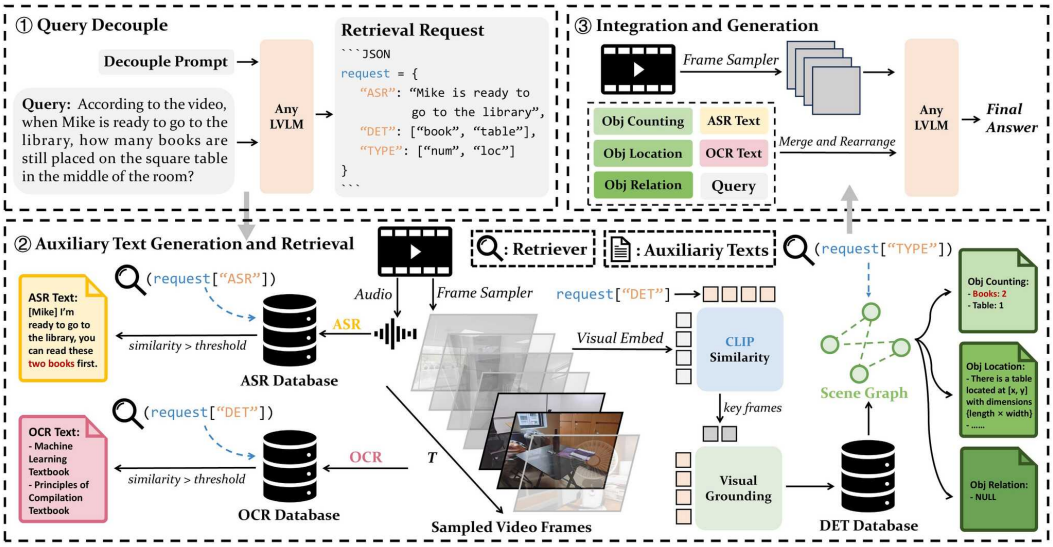

Figure 2: The framework of Video-RAG. Given a video and a user query, Video-RAG first decouples the query into a structured retrieval request, builds three modality-specific text databases offline, and retrieves relevant snippets at inference time. The retrieved texts are prepended to the final query for augmented LVLM inference.

GPT-4V. In contrast, Video-RAG augments the query with retrieved text in a single turn, requiring no extra visual tokens and no proprietary models.

Retrieval-Augmented Generation. RAG [13] has proven highly effective for knowledge-intensive NLP tasks by retrieving relevant passages from a corpus and prepending them to the input. Extensions to multimodal settings [1] typically retrieve images for image-QA tasks. OmAgent [38] applies tool-augmented LLMs to video understanding but requires multiple API calls. Our work applies RAG to the textual modalities within video—ASR, OCR, and object detections—enabling efficient single-turn augmentation.

Video Information Extraction. Prior work has explored individual modalities: ASR from speech [25], OCR from frames [9], and object detection [29]. VLog [19] combines BLIP2, CLIP, and Whisper to produce video-to-document transcriptions. VideoRAG [27, 10] builds external knowledge graphs from videos for retrieval. Our approach differs in that we perform *query-conditioned* retrieval of modality-specific texts, guided by a structured retrieval plan produced by the LVLM itself.

3 Method

3.1 Overview

Video-RAG augments any LVLM with a training-free retrieval stage that injects query-relevant, visually-aligned auxiliary texts into the prompt. As illustrated in Figure 2, the pipeline consists of three stages: (1) *offline database construction*, (2) *online query decomposition and retrieval*, and (3) *augmented LVLM inference*.

3.2 Offline Database Construction

For each video, we build three separate text databases corresponding to three complementary modalities:

ASR Database. We extract the audio track using `ffmpeg` and transcribe it with Whisper-large [25] in 30-second chunks. Each chunk produces a text segment indexed in chronological order, capturing the spoken content of the video.

OCR Database. We sample frames uniformly from the video and apply EasyOCR [9] to extract on-screen text. Text segments with confidence above 0.5 are retained; duplicate entries are removed. This captures text overlays, captions, signs, and other displayed text.

Object Detection Database. We use APE [29] (a generalist detection model) to detect objects in keyframes selected via CLIP-ViT-L/14-336 [24]. The keyframe selection is query-guided: we encode the user's object retrieval requests as CLIP text queries and select frames whose visual similarity to the query text exceeds a threshold β (normalized per-frame similarity). Detected objects with their bounding boxes are transformed into scene-graph descriptions using a structured text template that encodes spatial location, object count, and pairwise relations (e.g., "In the upper-left area: 2 blue balloons; above the long table"). This yields interpretable, visually-grounded scene descriptions that can be reasoned over by the LLM.

3.3 Query Decomposition

At inference time, given a user query q, Video-RAG first calls the LVLM in a *text-only* mode (no visual input) to decompose q into a structured retrieval request:

- **ASR**: A free-text description of what speech content might be relevant.
- **DET**: A list of physical entities to detect (up to 5), plus the desired information type (location / number / relation).
- **OCR**: OCR retrieval is guided by the DET entities (shared query).

This decomposition step adds a single text-only forward pass, avoiding any visual encoding overhead. The LVLM is prompted with in-context examples so that it outputs valid JSON, as shown below:

```
{"ASR": "The location and color of balloons, the number of blue
balloons.", "DET": ["blue balloons", "long table"], "TYPE":
["relation", "number"]}
```

3.4 Retrieval

Given the structured retrieval requests, we retrieve relevant snippets from each database using Contriever [8] with FAISS [12] (IndexFlatIP, inner-product similarity). We use *range search* with a threshold t: only documents whose similarity to the query exceeds t are included. This *threshold-based* strategy is crucial for efficiency: it avoids returning low-relevance documents, which waste context budget and can even degrade performance by introducing noise.

For ASR and OCR retrieval, the query is formed by combining the original user question q with the LVLM-generated ASR/DET retrieval request. Retrieved snippets are ordered chronologically. For object detection, CLIP-ViT-L/14-336 is used to score each frame against the DET query, and only frames above the CLIP threshold β are sent to APE for detection.

3.5 Augmented Inference

The retrieved texts are concatenated in the order **DET** $\rightarrow$ **ASR** $\rightarrow$ **OCR** and prepended to the final user query. This ordering places the most visually-specific information (object locations, counts, relations) closest to the question, followed by speech context, and finally text overlays. The augmented prompt is then processed by the LVLM together with the uniformly-sampled keyframes in the standard single-turn inference pipeline.

Prompt Format. The final query takes the form:

```
[Video has N frames in total, detected object info in specific
frames: Frame 3: ...]
[Video ASR information: ...]
[Video OCR information: ...]
Select the best answer ... Question: <question>
```

Table 1: Performance on the Video-MME [6] benchmark in without subtitles (w/o S), with subtitles (w/ S) and equipped with our Video-RAG (Ours), **Frames** and **Gain** means the input frame number and performance gain by applying Video-RAG compared to the baseline with subtitles. By applying Video-RAG to seven LVLMs, we observed an average performance improvement of 2.8% by adding only an average of ∼2.0K auxiliary texts compared to ∼3.0K full-subtitled tokens per sample. In particular, we perform better when applying Video-RAG with 72B LLaVA-Video [41] than the proprietary method GPT-4o [23] (77.4% vs. 77.2%). All results are our republication.

Model	Params	Frames	Short			Medium			Long			Overall			Gain
			w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	
Proprietary LVLMs															
GPT-4o [23]	-	384	80.0	82.8	-	70.3	76.6	-	65.3	72.1	-	71.9	77.2	-	-
Gemini-1.5-Pro [26]	-	0.5 fps	81.7	84.5	-	74.3	81.0	-	67.4	77.4	-	75.0	81.3	-	-
Open-Source LVLMs															
Video-LLaVA [17]	7B	8	45.3	46.1	49.5	38.0	40.7	43.0	36.2	38.1	42.5	39.9	41.6	45.0	+3.4
LLaVA-NeXT-Video [40]	7B	16	49.4	51.8	56.6	43.0	46.4	47.4	36.7	44.9	46.0	43.0	47.7	50.0	+2.3
VITA-1.5 [7]	7B	16	67.0	69.9	71.0	54.2	55.7	55.4	47.1	50.4	52.4	56.1	58.7	59.6	+0.9
LongVA [39]	7B	128	61.1	61.2	66.1	50.4	53.8	60.4	46.2	52.9	59.4	52.6	56.0	62.0	+6.0
Long-LLaVA [37]	7B	64	61.9	62.4	67.1	51.4	56.2	60.4	45.4	54.7	60.1	52.9	57.8	62.6	+4.8
Qwen2-VL [33]	72B	32	75.0	76.7	77.4	63.3	69.9	70.2	56.3	69.2	71.0	64.9	71.9	72.9	+1.0
LLaVA-Video [41]	72B	64	80.7	81.8	82.8	68.7	73.8	76.3	62.1	72.2	73.1	70.3	75.9	77.4	+1.5

Efficiency. The offline database construction is done once per video and cached. Online retrieval adds only ∼11 seconds per query (at t=0.3) and ∼1.9K tokens on a 7B model setup, which is significantly cheaper than adding ∼14 additional frames.

4 Experiments

4.1 Setup

Benchmarks. We evaluate on three long-video benchmarks:

- **Video-MME** [6]: A comprehensive multi-task benchmark with short (0–2 min), medium (4–15 min), and long (30–60 min) video categories. We report both with-subtitle (w/ S) and our Video-RAG results.
- **MLVU** [42]: A multiple-choice benchmark focused on long video understanding with diverse task types. We report the overall accuracy on the dev split.
- **LongVideoBench** [35]: A recently proposed benchmark targeting hour-long video comprehension. We report results on the validation set.

Base LVLMs. We evaluate Video-RAG as a plug-and-play augmentation on seven open-source LVLMs: Video-LLaVA [17], LLaVA-NeXT-Video [40], VITA-1.5 [7], LongVA [39], Long-LLaVA [37], Qwen2-VL-72B [33], and LLaVA-Video-72B [41].

Implementation Details. For all experiments, we sample frames uniformly (32–128 frames depending on the model). The retrieval threshold is set to t=0.3 and CLIP keyframe selection threshold to β=3.0 (normalized). ASR is extracted with Whisper-large [25], OCR with EasyOCR [9], and object detection with APE [29] using CLIP-ViT-L/14-336 [24] for keyframe selection. Retrieval is performed with Contriever [8] and FAISS [12].

4.2 Main Results on Video-MME

Table 1 presents Video-RAG performance on Video-MME across seven LVLMs. Compared to the subtitle baseline (w/ S), Video-RAG achieves an average gain of **+2.8%** across all models while adding only ∼2.0K auxiliary text tokens per sample—roughly equivalent to 14 extra frames in token cost.

Key observations:

- **Consistent gains**: All seven LVLMs benefit from Video-RAG, with gains ranging from +0.9% (VITA-1.5) to +6.0% (LongVA-7B). Models with larger context windows (LongVA, Long-LLaVA) benefit most, as they can better utilize the additional retrieved context.
- **Long video gains dominate**: The improvement is largest on the *Long* video category, where redundancy in uniform frame sampling is most severe and ASR/OCR signals are most informative.
- **Proprietary-level performance**: With LLaVA-Video-72B, Video-RAG achieves **77.4%** overall, surpassing GPT-4o (77.2%) and approaching Gemini-1.5-Pro (81.3% with subtitles). This demonstrates that open-source models can match proprietary performance when augmented with efficient, targeted auxiliary information.

4.3 Results on MLVU and LongVideoBench

Table 4 and Table 5 show results on MLVU and LongVideoBench (validation set).

Table 2: The overall performance in the multiple-choice task of the MLVU [42] benchmark. * donates the results of our replication.

Model	#Params	Frames	Overall
Proprietary LVLMs			
GPT-4o [23]	-	0.5 fps	64.6
Open-Source LVLMs			
VITA-1.5 [7]	7B	16	60.4
Video-CCAM [5]	14B	96	63.1
Video-XL [30]	7B	256	64.9
Aria [15]	25.3B	256	70.6
LLaVA-Video* [41]	7B	64	70.8
Oryx-1.5 [21]	32B	128	72.3
LLaVA-Video* [41]	72B	64	73.1
LLaVA-Video + Video-RAG	7B	64	72.4
LLaVA-Video + Video-RAG	72B	64	**73.8**

Table 3: The overall performance on the validation set of LongVideoBench [35]. * donates the results of our replication.

Model	#Params	Frames	Overall
Proprietary LVLMs			
Gemini-1.5-Pro [26]	-	256	64.0
GPT-4o [23]	-	256	**66.7**
Open-Source LVLMs			
VideoChat2-Mistral [16]	7B	8	39.3
ShareGPT4Video [2]	7B	8	39.7
LLaVA-Next-Mistral [20]	7B	8	49.1
PLLaVA [36]	34B	16	53.2
VITA-1.5 [7]	7B	16	53.6
LLaVA-Video* [41]	7B	64	56.6
LLaVA-Video* [41]	72B	64	61.9
LLaVA-Video + Video-RAG	7B	64	58.7
LLaVA-Video + Video-RAG	72B	64	65.4

On MLVU, Video-RAG improves LLaVA-Video-7B from 70.8% to **72.4%** (+1.6%) and LLaVA-Video-72B from 73.1% to **73.8%** (+0.7%), establishing a new state-of-the-art among open-source models. The 72B variant outperforms GPT-4o (64.6%) by a large margin.

On LongVideoBench (validation), LLaVA-Video-72B+Video-RAG achieves **65.4%**, surpassing Gemini-1.5-Pro (64.0%) and approaching GPT-4o (66.7%). The 7B variant also improves from 56.6% to 58.7%. These results confirm that Video-RAG generalizes well across diverse long-video benchmarks.

4.4 Ablation Studies

Effect of auxiliary text modalities. Table 4 (upper section) shows the contribution of each modality using Long-LLaVA-7B without RAG (naive concatenation). ASR alone already provides substantial gains, especially on long videos (+10.3% on Long). Adding OCR further improves Short and Medium categories. DET provides complementary gains on Medium and Long, particularly for spatial-relational queries. The combination of all three (DET+ASR+OCR) achieves the best overall score of 59.8%.

The lower section of Table 4 shows the same analysis *with* RAG-based retrieval (threshold t=0.3). The retrieval step substantially boosts all configurations: ASR+RAG improves from 57.4% to 61.6%, and the full DET+ASR+OCR+RAG achieves **62.1%** vs. 62.4% for the subtitle baseline w/ S, but at far lower token cost. Notably, RAG+ASR+OCR without DET achieves 62.0%, indicating that speech and text signals are the primary drivers.

Effect of retrieval threshold. Table 4 (right) shows the impact of varying the retrieval threshold t from 0.0 (all documents) to 1.0 (no documents). At t=0.3, we achieve the best overall accuracy (62.1%) with only 1.9K tokens and 11s per query. Lower thresholds (t=0.0) retrieve more tokens

Table 4: Results on combinations of different auxiliary texts in Video-MME [6] when using Long-LLaVA-7B [37] as the LVLM.

RAG	DET	OCR	ASR	Short	Medium	Long	Overall
				60.3	51.4	44.1	52.0
			✓	62.2	55.4	54.4	57.4
		✓	✓	64.0	56.2	55.0	58.4
	✓		✓	63.0	57.3	**56.4**	58.9
	✓	✓	✓	**64.3**	**58.8**	56.3	**59.8**
✓	✓			61.4	51.9	45.2	52.9
✓		✓		63.2	53.2	46.3	54.3
✓			✓	65.1	59.1	**60.7**	61.6
✓	✓	✓		64.1	54.6	48.4	55.7
✓	✓		✓	64.9	59.0	**60.7**	61.5
✓		✓	✓	66.3	**60.3**	59.3	62.0
✓	✓	✓	✓	**66.4**	60.2	59.8	**62.1**

Table 5: Performance with different thresholds of retrieval on Video-MME [6] when using Long-LLaVA-7B [37] as the LVLM. **#Token** and **Time** denote the total token number of the auxiliary texts and the average inference time per question, respectively.

t	#Token	Time	Short	Medium	Long	Overall
0.0	3.6K	36s	**67.6**	59.4	59.1	62.0
0.1	3.4K	30s	67.0	59.7	59.1	61.9
0.2	2.7K	18s	66.0	**60.2**	59.2	61.8
0.3	1.9K	11s	66.4	**60.2**	**59.8**	**62.1**
0.4	0.8K	8s	65.6	58.0	58.3	60.6
0.5	0.3K	7s	63.1	54.9	50.2	56.1
1.0	0.0K	6s	60.3	51.4	44.1	52.0
rnd	1.9K	11s	65.7	55.8	56.0	59.1

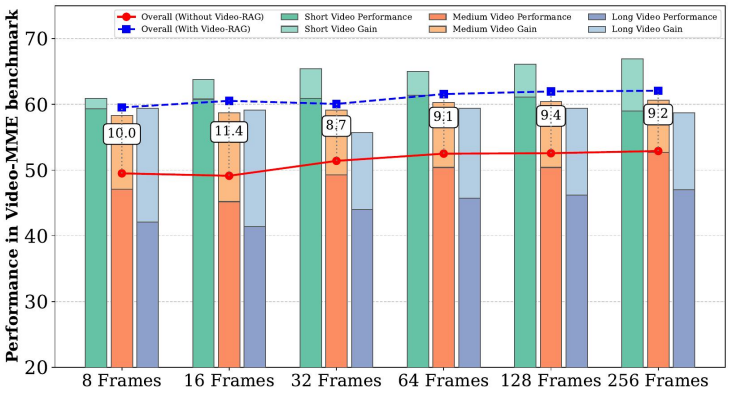

Figure 3: Performance gain with different frame sampling rates on Video-MME [6] when using LongVA-7B [39]. Video-RAG consistently outperforms the subtitle baseline, with larger gains at lower sampling rates.

(3.6K) but slightly reduce performance, suggesting that irrelevant context can hurt. Higher thresholds ($t \geq 0.4$) retrieve too few tokens, leading to performance degradation. We also compare against *random* retrieval of the same number of tokens (1.9K), which achieves only 59.1%—demonstrating that *query-relevance* is essential for the gains.

Effect of frame sampling rate. Figure 3 shows Video-RAG performance with different frame sampling rates on Video-MME using LongVA-7B. Video-RAG consistently outperforms the subtitle baseline across all frame rates, and the gains are particularly pronounced at lower frame rates (fewer frames), confirming that retrieved text effectively compensates for sparse visual coverage.

4.5 Qualitative Analysis

Qualitative example. Figure 4 shows a qualitative example on Video-MME where Video-RAG helps LLaVA-Video answer a question about specific on-screen text that is not visible in the sampled keyframes. The retrieved OCR snippet provides the critical missing information, enabling a correct answer that the baseline fails to produce.

Visualization of alignment. Figure 5 presents Grad-CAM [28] heatmaps of the last hidden state and t-SNE [32] visualizations of query and keyframe features. Without Video-RAG, the model's attention is diffuse across frames; with Video-RAG, attention concentrates on the most relevant keyframes, and the t-SNE plot shows the query embedding moves closer to the informative keyframes. This confirms that Video-RAG improves cross-modal alignment between the query and the retrieved visual content.

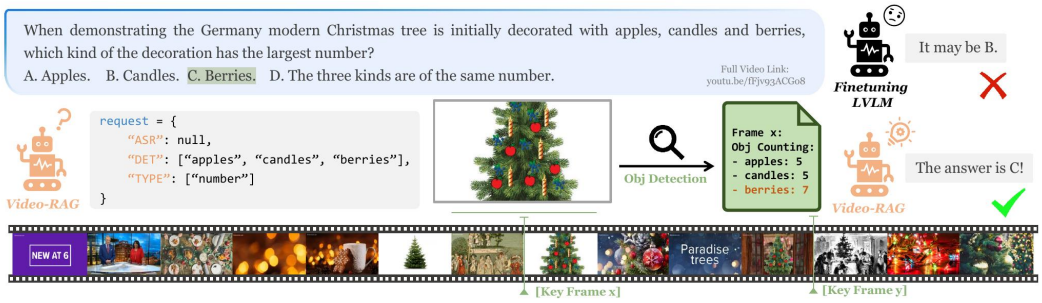

Figure 4: Qualitative result on Video-MME [6] when applying Video-RAG with LLaVA-Video [41]. The retrieved auxiliary texts (OCR/ASR) provide crucial information not visible in the sampled frames, leading to a correct answer.

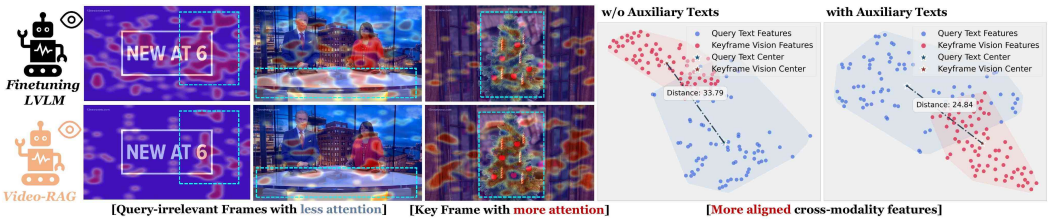

Figure 5: Grad-CAM visualizations of the last hidden state heatmap and t-SNE visualizations of query and keyframe features. Video-RAG improves query–keyframe alignment, as shown by both the attention heatmaps and the t-SNE embeddings.

5 Conclusion

We presented **Video-RAG**, a training-free, plug-and-play retrieval-augmented generation pipeline for long video comprehension. Video-RAG augments any LVLM with query-relevant, visually-aligned auxiliary texts—extracted via ASR (Whisper), OCR (EasyOCR), and object detection (APE)—retrieved using Contriever+FAISS with a threshold-based range search. Through a single-turn query decomposition and parallel retrieval stage, Video-RAG injects only ∼2.0K extra tokens per sample while delivering consistent performance gains across seven open-source LVLMs on Video-MME (+2.8% avg), MLVU, and LongVideoBench. With LLaVA-Video-72B, Video-RAG achieves proprietary-level performance, surpassing GPT-4o on Video-MME (77.4% vs. 77.2%) and Gemini-1.5-Pro on LongVideoBench (65.4% vs. 64.0%).

Video-RAG demonstrates that injecting modality-specific, query-conditioned text into the LVLM prompt is a simple yet surprisingly effective strategy for long video understanding—one that is complementary to advances in model architecture and training. We hope this work spurs further research into efficient, training-free augmentation of multimodal models.

Limitations and Future Work. Video-RAG has several limitations worth noting. The offline preprocessing introduces a one-time construction cost per video that may be prohibitive for very large collections. Retrieval quality also depends on the accuracy of the underlying tools: ASR transcription errors or OCR failures can propagate to the LVLM and potentially mislead it. The threshold t=0.3 is globally fixed and may benefit from per-query adaptive calibration to better balance recall and precision. Looking ahead, we plan to investigate joint cross-modal retrieval that considers visual and textual features simultaneously, adaptive frame sampling steered by retrieved signals, and extending Video-RAG to open-ended generation tasks beyond multiple-choice QA.

Acknowledgments

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